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AGE 418
Sensors and Controls in Precision Agriculture

Lecture 2 Compendium:
GIS, GPS, and GNSS Basics

The Spatial Framework of Modern Farming

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1 1. Introduction: The "Where" in Agriculture

In Lecture 1, we established that Precision Agriculture (PA) manages variability. To manage variability, we must first locate it.

Before we can use sensors to measure soil moisture or cameras to detect weeds, we need a spatial framework to answer two fundamental questions:

1. **Where** am I? (Positioning - GNSS)
2. **What** is around me? (Mapping - GIS)

This lecture covers the technologies that create the digital map of the farm.

2 2. Geographic Information Systems (GIS)

2.1 Definition

A Geographic Information System (GIS) is a computer-based system designed to capture, store, manipulate, analyze, manage, and present all types of spatial or geographical data. In agriculture, GIS is the "brain" that organizes data layers.

The Layer Cake Concept

GIS treats the real world as a stack of thematic layers.

- **Layer 1:** Topography (Elevation)
- **Layer 2:** Soil Type (Clay/Sand)
- **Layer 3:** Yield History (Last season's harvest)
- **Layer 4:** Current Crop Health (NDVI)

Analysis happens where these layers intersect vertically. For example: "Show me areas where Yield is Low AND Soil is Sandy."

2.2 Data Models: Vector vs. Raster

Agricultural data comes in two distinct formats. Engineers must understand the difference to process data correctly.

2.2.1 1. Vector Data (Discrete)

Vector data uses mathematical coordinates (X, Y) to represent discrete objects. It is scalable without loss of quality.

- **Points:** A soil sampling location, a well, or a specific tree.
- **Lines:** An irrigation pipe, a road, or a tractor guidance path (A-B line).
- **Polygons:** A field boundary, a soil zone, or a pond.

Standard Format: The Shapefile (.shp).

2.2.2 2. Raster Data (Continuous)

Raster data represents the world as a grid of cells (pixels). Each cell has a single value.

- Used for continuous variables where boundaries are not sharp.
- **Examples:** Satellite imagery, Digital Elevation Models (DEM), Yield Maps.
- **Resolution:** Defined by pixel size (e.g., 10m x 10m pixel).

Standard Format: GeoTIFF (.tif).

Table 1: Comparison of Vector and Raster Data in Agriculture

Feature	Vector Data	Raster Data
Structure	Points, Lines, Polygons	Grid of Pixels
Precision	Very High (Exact coordinates)	Limited by pixel size
File Size	Small	Large
Best For	Field Boundaries, Roads	Imagery, Elevation, Yield
Processing	Topology analysis	Image processing, Map algebra

3 3. Coordinate Systems & Projections

A common error in agricultural engineering is mixing up coordinate systems, leading to "floating fields" or incorrect area calculations.

3.1 Geographic Coordinate Systems (GCS)

- **Model:** A 3D sphere (or ellipsoid).
- **Units:** Degrees (Latitude, Longitude).
- **Standard: WGS84** (World Geodetic System 1984). This is the native language of GPS receivers.
- *Use:* Storing raw data.

3.2 Projected Coordinate Systems (PCS)

- **Model:** A flat 2D plane.
- **Units:** Meters.
- **Standard: UTM** (Universal Transverse Mercator).
- *Use:* Calculating area (hectares) and distance (meters). You cannot calculate hectares accurately using Degrees.

Tanzania's UTM Zones

Tanzania spans two primary UTM zones:

- **Zone 36S:** Western Tanzania (Mwanza, Tabora, Mbeya).
- **Zone 37S:** Eastern Tanzania (Dar es Salaam, Morogoro, Kilimanjaro).

*Note: When setting up a farm GIS in Morogoro, always project data to **WGS84 / UTM Zone 37S**.*

4 4. GNSS: Global Navigation Satellite Systems

While commonly called "GPS," the correct engineering term is **GNSS**.

4.1 The Constellations

GNSS refers to the collection of all satellite navigation systems:

1. **GPS (NAVSTAR):** United States (approx. 31 satellites).
2. **GLONASS:** Russia (approx. 24 satellites).
3. **Galileo:** European Union (High precision for civilians).
4. **BeiDou:** China.

Modern agricultural receivers track all four constellations simultaneously to ensure signal availability even near trees or hills.

4.2 How Positioning Works: Trilateration

GNSS receivers do not measure distance directly; they measure **Time**.

1. The satellite sends a signal containing the exact time it left the satellite.
2. The receiver logs the exact time the signal arrived.
3. **Difference in Time (Δt) $\times$ Speed of Light (c) = Distance.**

$$Distance = \Delta t \times 299,792,458 \text{ m/s} \tag{1}$$

By calculating the distance to at least **4 satellites**, the receiver can mathematically solve for X (Latitude), Y (Longitude), Z (Elevation), and Time Error.

5 5. Sources of Error & Accuracy

A standard smartphone GPS is accurate to roughly **5-10 meters**. This is useless for a tractor trying to plant rows 75cm apart. Why is it inaccurate?

5.1 Major Error Sources

1. **Ionospheric Delay:** Charged particles in the upper atmosphere slow down the signal, messing up the "Time = Distance" calculation.
2. **Multipath:** Signals bounce off trees, metal roofs, or hills before hitting the receiver, making the distance appear longer than it is.
3. **Clock Errors:** Tiny discrepancies between satellite atomic clocks and receiver quartz clocks.
4. **Satellite Geometry (DOP):** If all visible satellites are clustered in one part of the sky, triangulation is weak (High Dilution of Precision).

5.2 Correction Levels

- **Autonomous (Standalone):** Accuracy $\approx$ 5-10m. (Phone GPS). *Use: Scouting.*
- **DGPS (Differential GPS):** Accuracy $\approx$ sub-meter (30-50cm). Uses correction from a reference station. *Use: Spraying, Spreading.*
- **RTK (Real-Time Kinematic):** Accuracy $\approx$ 2.5cm. Uses a local base station and carrier-phase tracking. *Use: Planting, Drip Tape Installation, weeding.*

6 6. Differential Correction: The Key to Precision

To achieve high accuracy, we use **Differential Correction**.

6.1 The Concept

1. We place a **Base Station** at a known, fixed location (Surveyed point). 2. The Base Station receives signals from satellites. It knows exactly where it *is*, so it calculates the error in the satellite signal (e.g., "Satellite 12 is reporting I am 5 meters North, so there is a 5-meter error"). 3. The Base sends this **Correction Factor** to the tractor (**Rover**) via Radio or Internet. 4. The Tractor applies the correction to its own reading.

6.2 RTK Delivery Methods

- **Radio RTK:** You own the base station. It sits on a tripod at the field edge. Range: 10-15km (Line of Sight).
- **NTRIP (Network RTK):** Corrections are sent over the **Mobile Internet** (Cellular). You don't own a base; you subscribe to a network (like a CORS network).

Challenge in Tanzania

In many rural parts of Tanzania, NTRIP is difficult due to poor cellular network coverage. Therefore, Radio RTK (Local Base Station) is often the preferred engineering solution for large estates (e.g., Mtibwa Sugar).

7 7. Applications in PA

7.1 1. Yield Monitoring

A combine harvester records the mass of grain and the GPS location every second. This creates a map showing high and low-yield zones.

7.2 2. Auto-Steer (Guidance)

The most common application. It reduces overlap (double spraying) and skips.

- **Parallel Tracking:** Straight lines (A-B lines).
- **Contour Tracking:** Following curved terraces (common in Iringa/Mbeya).

7.3 3. Variable Rate Application (VRA)

The GPS tells the tractor computer "You are in Zone A (Sandy Soil)." The computer tells the sprayer "Reduce rate to 50%."

8 8. Lecture Assessment

8.1 Part A: Simple Concept Questions

1. Explain the difference between **GPS** and **GNSS**. Why is a GNSS receiver preferred over a GPS-only receiver in a field with tree lines?

2. Define **GIS** (Geographic Information System) in one sentence.

3. Explain the "Layer Cake" concept in GIS with agricultural examples.

4. Differentiate between **Vector** and **Raster** data types. Which one is best for representing a field boundary?

5. What is the standard projected coordinate system used for mapping in Eastern Tanzania (e.g., Morogoro)?

8.2 Part B: Advanced Analysis & Real-World Issues

1. **Trilateration Physics:** Explain the mathematical principle of trilateration. Why are at least 4 satellites required for an accurate 3D position fix (Latitude, Longitude, Altitude), rather than just 3?

2. **NBS Shapefiles Analysis:** Visit the **National Bureau of Statistics (NBS) Tanzania** website or their open data portal. *Task:* Download the "2012 Census Ward Shapefiles." Explain the file structure of a shapefile (shp, shx, dbf, prj). If you lose the '.prj' file, what happens to your map when you load it into QGIS?

3. **RCMRD Geoportal Exploration:** Visit the **RCMRD** (Regional Centre for Mapping of Resources for Development) Geoportal. *Task:* Find the "Land Cover Map" for Tanzania. Identify an area classified as "Cropland" near Arusha. Explain whether this dataset is Vector or Raster, and determine its spatial resolution. Is this resolution sufficient for precision weeding on a 2-acre farm?

4. **CORS Network Planning:** Assume you are an engineer designing an RTK network for the **SAGCOT** (Southern Agricultural Growth Corridor of Tanzania) region. *Task:* Using Google Earth, plot 3 hypothetical locations for CORS (Continuously Operating Reference Stations) towers between Morogoro and Iringa. Justify your locations based on topography (Line of Sight) and power/internet availability.

5. **Sentinel-2 Band Analysis:** Download a Sentinel-2 satellite image tile for **Mbarali Rice Farms** using **Copernicus Open Access Hub** or **GloVis**. *Task:* Explain the difference between Band 4 (Red) and Band 8 (Near Infrared). Which of these bands would look brighter (higher pixel value) over a healthy rice paddy, and why?

6. **RTK vs. DGPS Economics:** A sugarcane estate needs to buy guidance systems for 10 tractors. *Task:* Tractor A does Tillage (needs +/- 30cm accuracy). Tractor B does Planting (needs +/- 2.5cm accuracy). Recommend the correction signal type (Autonomous, DGPS, or RTK) for each tractor and justify the cost difference.

7. **OpenStreetMap (OSM) for Logistics:** Go to **OpenStreetMap.org** and zoom into a rural farming area in Tanzania (e.g., **Mafia Island** coconut plantations). *Task:* Compare the road network data in OSM versus Google Maps. As an engineer designing an autonomous tractor logistics algorithm, why might the vector data from OSM be more useful than the raster background of Google Maps?

8. **Multipath Error Mitigation:** You are operating a GNSS receiver in a dense mango orchard in Tanga. The position keeps jumping by 5 meters. *Task:* Explain the phenomenon of "Multipath Error" in this context. What setting on the receiver (Elevation Mask) could you adjust to ignore satellites low on the horizon that are likely blocked by trees?

9. **Coordinate Conversion:** A drone takes a photo at Lat: -6.848, Long: 37.662 (WGS84). To calculate the area of the field in hectares, you need to project this to UTM Zone 37S. *Task:* Explain why calculating area using Degrees (Lat/Long) yields incorrect results compared to Meters (UTM).

10. **Visualizing Elevation (SRTM):** Download SRTM (Shuttle Radar Topography Mission) data for the Uluguru Mountains. *Task:* This is Raster data. How would you convert this into Vector contour lines (e.g., 10-meter intervals) to help a farmer plan terracing for soil conservation?